

Physical Chemistry–III (CHB-504)

Unit 4: Photochemistry

Satyam Kumar Singh

Ye to ratta maarne ki cheez hai

Contents

1	Fundamental Laws of Photochemistry	3
1.1	Grotthuss–Draper Law (First Law of Photochemistry)	3
1.2	Stark–Einstein Law of Photochemical Equivalence (Second Law)	3
1.3	Energy of One Einstein	3
1.4	Why Many Reactions Do Not Obey the Stark–Einstein Law	3
2	Quantum Yield	4
2.1	Definition	4
2.2	Primary vs. Overall Quantum Yield	4
2.3	Worked Example: Quantum Yield Calculation	4
3	Jablonski Diagram and Photophysical Processes	4
3.1	Jablonski Diagram	5
3.2	Photophysical Processes Defined	5
3.3	Thermal Equilibrium vs. Photostationary State	5
3.4	Resonance Emission	6
4	Fluorescence Quenching and Stern–Volmer Equation	6
4.1	Mechanism of Fluorescence Quenching	6
4.2	Derivation of the Stern–Volmer Equation	6
4.3	Stern–Volmer Plot and Determination of Constants	7
5	Photochemical Reaction: $\text{H}_2 + \text{Br}_2$	7
5.1	Mechanism (Chain Reaction)	7
5.2	Derivation of Rate of HBr Formation	8
5.3	Quantum Yield of HBr Formation	8
6	Photochemical Decomposition of HI	9
6.1	Mechanism	9
6.2	Quantum Yield of HI Decomposition	9

7	Dimerisation of Anthracene: Quantum Yield	9
7.1	Mechanism	9
7.2	Steady-State on A^*	9
7.3	Quantum Yield of Dimerisation	9
7.4	Why $\Phi(A_2)$ is Small and Concentration-Dependent	10
8	General Photostationary-State Kinetics	10
8.1	Three Competing Deactivation Pathways	10
9	Chemical Actinometry	10
9.1	Definition	10
9.2	Two Important Actinometers	11
9.2.1	1. Uranyl Oxalate Actinometer	11
9.2.2	2. Ferrioxalate Actinometer (Parker)	11
9.3	Procedure	11
10	Reasons for Low and High Quantum Yield	11
10.1	Low Quantum Yield ($\Phi \ll 1$)	11
10.2	High Quantum Yield ($\Phi \gg 1$)	11
11	Summary and Key Formulae	12

Fundamental Laws of Photochemistry

Grotthuss–Draper Law (First Law of Photochemistry)

Only the radiation that is **absorbed** by a substance can produce a photochemical change. Radiation that is transmitted or reflected has no photochemical effect.

This is the *qualitative* law: absorption is a necessary (but not sufficient) condition for photochemical reaction.

Stark–Einstein Law of Photochemical Equivalence (Second Law)

Each molecule that participates in a photochemical reaction absorbs **one** quantum (photon) of radiation that causes the reaction. Each photon activates exactly one molecule.

One mole of photons is called one **Einstein**. Its energy is:

$$E_{\text{Einstein}} = N_A h \nu = \frac{N_A h c}{\lambda}$$

where $h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^8$ m/s, λ = wavelength (m).

Energy of One Einstein

$$E_{\text{Einstein}} = \frac{N_A h c}{\lambda} = \frac{(6.022 \times 10^{23})(6.626 \times 10^{-34})(3 \times 10^8)}{\lambda(\text{m})} = \frac{0.11962}{\lambda(\text{m})} \text{ J mol}^{-1}$$

Example: For $\lambda = 500 \text{ nm} = 5 \times 10^{-7} \text{ m}$:

$$E = \frac{0.11962}{5 \times 10^{-7}} = 2.392 \times 10^5 \text{ J mol}^{-1} = \boxed{239.2 \text{ kJ mol}^{-1}}$$

Why Many Reactions Do Not Obey the Stark–Einstein Law

The quantum yield (defined in Section 2) often differs from unity because:

1. **Chain reactions:** One absorbed photon initiates a long radical chain, so very many molecules react per photon ($\Phi \gg 1$; e.g., $\text{H}_2 + \text{Cl}_2$, $\Phi \sim 10^6$).
2. **Competing deactivation:** The excited molecule may lose energy by fluorescence, phosphorescence, internal conversion (IC), intersystem crossing (ISC), or collisional quenching before reacting ($\Phi < 1$).
3. **Back reactions:** Products recombine, reducing net chemical change ($\Phi < 1$).
4. **Quenching by solvent/impurities:** Energy transferred to surroundings ($\Phi < 1$).
5. **Energy pooling (sensitisation):** Two excited molecules exchange energy without net chemical change ($\Phi < 1$).

Quantum Yield

Definition

$$\Phi = \frac{\text{number of molecules reacting (or product formed)}}{\text{number of photons absorbed}} = \frac{\text{moles of reactant consumed}}{\text{moles of photons (Einsteins) absorbed}}$$

Primary vs. Overall Quantum Yield

- **Primary quantum yield** Φ_{prim} : molecules undergoing the primary photochemical act (bond cleavage, ionisation, *etc.*) per photon absorbed. By the Stark–Einstein law, $\Phi_{\text{prim}} \leq 1$.
- **Overall quantum yield** Φ_{overall} : accounts for all subsequent thermal (dark) reactions. Can be $\ll 1$ or $\gg 1$.

Key distinction: Each photon can activate only one molecule (Stark–Einstein), so $\Phi_{\text{prim}} \leq 1$. But a radical chain propagated by that one activation event can consume thousands of molecules, giving $\Phi_{\text{overall}} \gg 1$.

Worked Example: Quantum Yield Calculation

Problem: 0.002 mol of a substance reacts in 20 min 4 s during which 2.0×10^6 photons per second are absorbed. ($N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$.)

Step 1: Total time = $20 \times 60 + 4 = 1204 \text{ s}$.

Step 2: Total photons = $2.0 \times 10^6 \times 1204 = 2.408 \times 10^9$.

Step 3: Moles of photons = $\frac{2.408 \times 10^9}{6.02 \times 10^{23}} = 4.0 \times 10^{-15} \text{ mol (Einstein)}$.

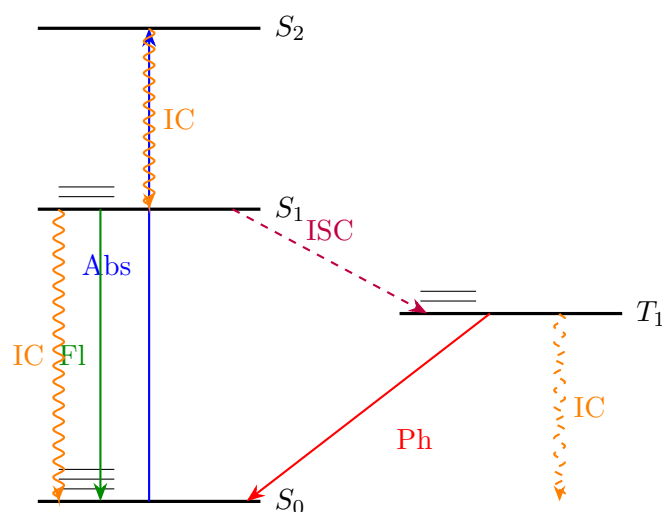
Step 4:

$$\Phi = \frac{0.002}{4.0 \times 10^{-15}} = \boxed{5.0 \times 10^{11}}$$

The extremely large Φ is characteristic of a radical-chain reaction.

Jablonski Diagram and Photophysical Processes

Jablonski Diagram



Abs=Absorption **Fl**=Fluorescence **Ph**=Phosphorescence
IC=Internal Conversion **ISC**=Intersystem Crossing

Photophysical Processes Defined

Process	Description
Absorption	$S_0 + h\nu \rightarrow S_n$ ($n = 1, 2, \dots$). Timescale $\sim 10^{-15}$ s.
Internal Conversion (IC)	Radiationless transition between states of the <i>same</i> spin multiplicity (e.g. $S_2 \rightarrow S_1$, $S_1 \rightarrow S_0$). Energy released as molecular vibration/heat.
Intersystem Crossing (ISC)	Radiationless transition between states of <i>different</i> spin multiplicity ($S_1 \rightarrow T_1$). Requires spin-orbit coupling.
Fluorescence	Radiative $S_1 \rightarrow S_0$ emission. Same spin multiplicity (allowed), fast ($\sim 10^{-9}$ s). Ceases immediately when irradiation stops.
Phosphorescence	Radiative $T_1 \rightarrow S_0$ emission. Different spin (spin-forbidden, slow: 10^{-3} – 10^2 s). Persists after light source removed (afterglow).
Quenching	Collisional deactivation of A^* by quencher Q; energy dissipated without radiation or reaction.

Thermal Equilibrium vs. Photostationary State

Thermal equilibrium: The system obeys the Boltzmann distribution over all energy levels; temperature alone determines level populations; no net flows of energy or matter.

Photostationary state: A dynamic *steady state* sustained by *continuous* radiation — the rate of population of the excited state (absorption) exactly equals the rate of depopulation (all deactivation channels combined). When the light is switched off, the system relaxes back to thermal equilibrium.

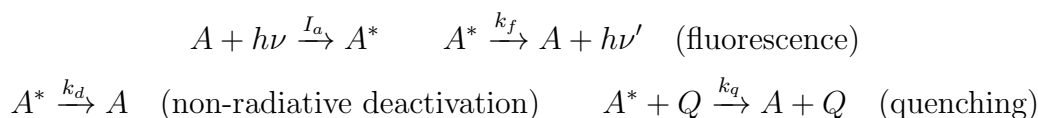
Feature	Thermal Equilibrium	Photostationary State
Driving force	None	Continuous irradiation
Population law	Boltzmann	Non-Boltzmann
Persistence	Permanent	Only while light is on
Temperature role	Defines all populations	Irrelevant to $[A^*]$
Example	Gas in dark box	Dye under lamp

Resonance Emission

Resonance emission (resonance fluorescence) is the re-emission of radiation at *exactly the same frequency* as the absorbed radiation ($\nu_{\text{emitted}} = \nu_{\text{absorbed}}$). There is no vibrational relaxation between absorption and emission. It is typically observed in *atomic* vapours (e.g. sodium D-line) where there are no vibrational sub-levels. In molecules, vibrational relaxation in S_1 occurs first, so emitted fluorescence is red-shifted (Stokes shift), and resonance emission is usually absent.

Fluorescence Quenching and Stern–Volmer Equation

Mechanism of Fluorescence Quenching



Derivation of the Stern–Volmer Equation

Apply the **photostationary-state approximation** to $[A^*]$:

$$\frac{d[A^*]}{dt} = I_a - k_f[A^*] - k_d[A^*] - k_q[Q][A^*] = 0$$

$$[A^*] = \frac{I_a}{k_f + k_d + k_q[Q]}$$

Fluorescence intensity $I_f \propto k_f[A^*]$:

$$I_f = \frac{k_f I_a}{k_f + k_d + k_q[Q]}$$

Without quencher ($[Q] = 0$):

$$I_f^0 = \frac{k_f I_a}{k_f + k_d}$$

Dividing:

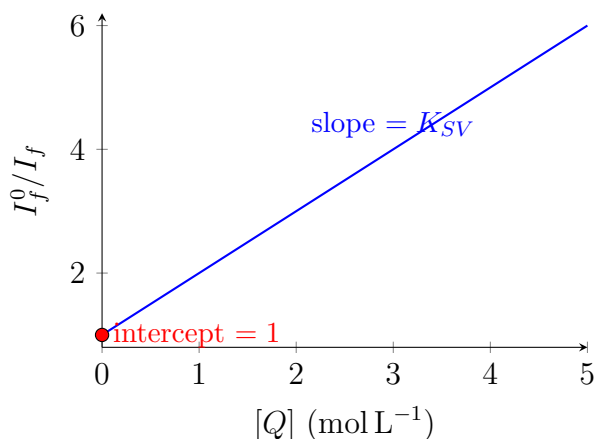
$$\frac{I_f^0}{I_f} = \frac{k_f + k_d + k_q[Q]}{k_f + k_d} = 1 + \frac{k_q[Q]}{k_f + k_d}$$

The natural (unquenched) fluorescence lifetime is $\tau_0 = 1/(k_f + k_d)$, so:

$$\boxed{\frac{I_f^0}{I_f} = 1 + k_q\tau_0[Q] = 1 + K_{SV}[Q]}$$

where $K_{SV} = k_q\tau_0$ is the **Stern–Volmer constant**.

Stern–Volmer Plot and Determination of Constants

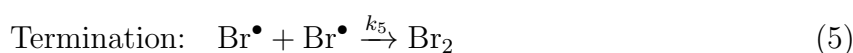
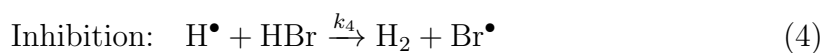
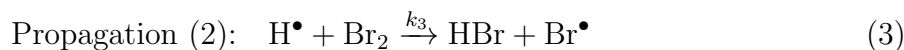
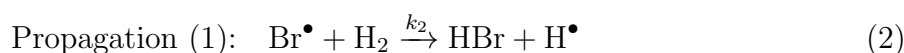


From the Stern–Volmer plot:

- **Slope** = $K_{SV} = k_q\tau_0 \rightarrow$ if τ_0 is known independently, $k_q = K_{SV}/\tau_0$.
- **Intercept** = 1 (confirms the mechanism).
- k_f can be determined separately from $\tau_0 = 1/(k_f + k_d)$ and the fluorescence quantum yield $\Phi_f = k_f/(k_f + k_d)$, giving $k_f = \Phi_f/\tau_0$.

Photochemical Reaction: $\text{H}_2 + \text{Br}_2$

Mechanism (Chain Reaction)



Derivation of Rate of HBr Formation

Net rate of HBr formation (from steps 2, 3, 4):

$$\frac{d[\text{HBr}]}{dt} = k_2[\text{Br}^\bullet][\text{H}_2] + k_3[\text{H}^\bullet][\text{Br}_2] - k_4[\text{H}^\bullet][\text{HBr}]$$

Steady-state on $\text{H}^\bullet$ ($d[\text{H}^\bullet]/dt = 0$):

$$k_2[\text{Br}^\bullet][\text{H}_2] - k_3[\text{H}^\bullet][\text{Br}_2] - k_4[\text{H}^\bullet][\text{HBr}] = 0$$

$$[\text{H}^\bullet] = \frac{k_2[\text{Br}^\bullet][\text{H}_2]}{k_3[\text{Br}_2] + k_4[\text{HBr}]}$$

Steady-state on $\text{Br}^\bullet$ ($d[\text{Br}^\bullet]/dt = 0$):

The initiation rate equals twice the termination rate (each Br_2 photolysis gives 2 $\text{Br}^\bullet$, each termination removes 2 $\text{Br}^\bullet$):

$$2k_1I_a - 2k_5[\text{Br}^\bullet]^2 = 0 \quad \Rightarrow \quad [\text{Br}^\bullet] = \left(\frac{k_1I_a}{k_5} \right)^{1/2}$$

where I_a is the rate of photon absorption ($\text{Einstein L}^{-1}\text{s}^{-1}$).

Substituting $[\text{Br}^\bullet]$ and $[\text{H}^\bullet]$ into the HBr rate:

$$\frac{d[\text{HBr}]}{dt} = 2k_2[\text{Br}^\bullet][\text{H}_2] \cdot \frac{k_3[\text{Br}_2]}{k_3[\text{Br}_2] + k_4[\text{HBr}]} + k_2[\text{Br}^\bullet][\text{H}_2] - k_2[\text{Br}^\bullet][\text{H}_2]$$

After careful algebra (adding contributions from both propagation steps and cancelling the inhibition term):

$$\boxed{\frac{d[\text{HBr}]}{dt} = \frac{2k_2 \left(\frac{k_1}{k_5} \right)^{1/2} I_a^{1/2} [\text{H}_2][\text{Br}_2]^{1/2}}{1 + \frac{k_4[\text{HBr}]}{k_3[\text{Br}_2]}} = \frac{k [\text{H}_2][\text{Br}_2]^{1/2}}{1 + k'[\text{HBr}]/[\text{Br}_2]}}$$

where $k = 2k_2(k_1/k_5)^{1/2}I_a^{1/2}$ and $k' = k_4/k_3$.

Quantum Yield of HBr Formation

$$\Phi(\text{HBr}) = \frac{d[\text{HBr}]/dt}{I_a} = \frac{2k_2(k_1/k_5)^{1/2}I_a^{-1/2}[\text{H}_2][\text{Br}_2]^{1/2}}{1 + k'[\text{HBr}]/[\text{Br}_2]}$$

At the start of the reaction ($[\text{HBr}] = 0$) and noting $I_a^{-1/2}$ dependence:

$$\Phi_{\text{initial}}(\text{HBr}) \gg 1 \quad (\text{typically } 10^4\text{--}10^6)$$

Each absorbed photon generates one $\text{Br}^\bullet$; this initiates a chain in which each propagation cycle produces two HBr molecules, so the overall yield is enormous. The inhibition step (step 4) explains why HBr, once formed, slows the reaction (negative feedback via the denominator term $k'[\text{HBr}]/[\text{Br}_2]$).

Photochemical Decomposition of HI

Mechanism



Overall: $2 \text{HI} \xrightarrow{h\nu} \text{H}_2 + \text{I}_2$.

Quantum Yield of HI Decomposition

Per photon absorbed: step 1 consumes one HI; step 2 consumes another HI. Therefore **two** HI molecules are decomposed per photon:

$$\boxed{\Phi(\text{HI decomp}) = 2}$$

Explanation: The primary quantum yield is 1 (Stark–Einstein), but the $\text{H}^\bullet$ radical immediately reacts with a second HI molecule in a fast thermal step, doubling the net consumption. This is a *short chain* (chain length = 1 thermal step), not a long chain.

Dimerisation of Anthracene: Quantum Yield

Mechanism

The UV-induced dimerisation of anthracene A in benzene solution:



Steady-State on A^*

$$\frac{d[A^*]}{dt} = I_a - k_f[A^*] - k_2[A^*][A] = 0 \quad \Longrightarrow \quad [A^*] = \frac{I_a}{k_f + k_2[A]}$$

Quantum Yield of Dimerisation

$$\frac{d[A_2]}{dt} = k_2[A^*][A] = \frac{k_2 I_a [A]}{k_f + k_2[A]}$$

$$\boxed{\Phi(A_2) = \frac{d[A_2]/dt}{I_a} = \frac{k_2[A]}{k_f + k_2[A]}}$$

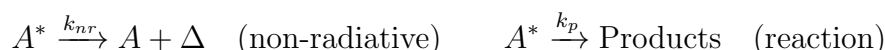
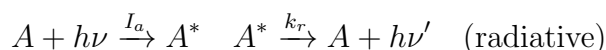
Why $\Phi(A_2)$ is Small and Concentration-Dependent

- Anthracene has a very large fluorescence rate constant k_f ; most A^* deactivates radiatively before encountering another A molecule.
- Since $k_f \gg k_2[A]$ at ordinary concentrations: $\Phi(A_2) \approx k_2[A]/k_f \ll 1$.
- **As $[A]$ increases**, $\Phi(A_2)$ increases because the bimolecular quenching (dimerisation) path competes more effectively with fluorescence.
- The thermal back-dissociation ($A_2 \rightarrow 2A$) in the dark further reduces the net dimer yield.
- The concentration dependence itself *confirms* the bimolecular mechanism involving A^* reacting with a ground-state A molecule.

General Photostationary-State Kinetics

Three Competing Deactivation Pathways

For any photochemically excited molecule A^* :



Steady-state on $[A^*]$:

$$[A^*]_{ss} = \frac{I_a}{k_r + k_{nr} + k_p}$$

Rate of product formation:

$$\frac{d[P]}{dt} = \frac{k_p I_a}{k_r + k_{nr} + k_p}$$

Overall quantum yield:

$$\Phi = \frac{k_p}{k_r + k_{nr} + k_p}$$

This shows $\Phi < 1$ whenever radiative or non-radiative deactivation competes with reaction.

Chemical Actinometry

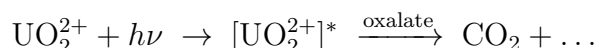
Definition

Chemical actinometry uses a photochemical reaction of *known* quantum yield Φ_{act} to determine the intensity I_a (photons absorbed per unit time) of a light beam:

$$I_a = \frac{\text{rate of reaction (mol L}^{-1}\text{s}^{-1})}{\Phi_{\text{act}}}$$

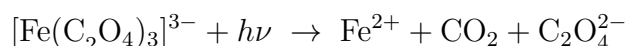
Two Important Actinometers

1. Uranyl Oxalate Actinometer



The sensitised decomposition of oxalate is measured by back-titration with KMnO_4 . $\Phi \approx 0.5\text{--}0.6$ (wavelength-dependent, 200–400 nm range).

2. Ferrioxalate Actinometer (Parker)



Fe^{2+} produced is measured spectrophotometrically with 1,10-phenanthroline (orange complex at 510 nm). $\Phi \approx 1.0\text{--}1.2$; applicable from 250–500 nm; most widely used modern actinometer.

Procedure

1. Expose actinometer solution to light for time t under identical conditions to the target reaction.
2. Measure amount of product formed ($\Delta n_{\text{product}}$, mol L⁻¹).
3. Calculate $I_a = \Delta n_{\text{product}} / (t \Phi_{\text{act}})$ (Einstein L⁻¹s⁻¹).

Reasons for Low and High Quantum Yield

Low Quantum Yield ($\Phi \ll 1$)

1. Competing IC, ISC, fluorescence, phosphorescence deactivate A^* without reaction.
2. Collisional quenching by solvent molecules or impurities.
3. Rapid back reaction: products immediately re-form reactants (e.g., anthracene dimer dissociates thermally).
4. Very short excited-state lifetime: molecule deactivates before a reactive collision.
5. Energy pooling: two excited molecules exchange energy, leaving both inactive.

High Quantum Yield ($\Phi \gg 1$)

1. **Radical chain reactions:** each photon initiates a chain of many elementary steps (e.g., $\text{H}_2 + \text{Cl}_2$, $\Phi \sim 10^6$).
2. **Photosensitisation:** excited sensitiser transfers energy to many substrate molecules sequentially.
3. **Cascade reactions:** each primary product undergoes rapid further thermal reactions.

Summary and Key Formulae

Quantity / Law	Expression
Stark–Einstein law	One photon activates one molecule
Energy of one Einstein	$E = N_A h c / \lambda$
Quantum yield (definition)	$\Phi = (\text{molecules reacted}) / (\text{photons absorbed})$
Photostationary $[A^*]$	$[A^*]_{ss} = I_a / (k_r + k_{nr} + k_p)$
General Φ	$\Phi = k_p / (k_r + k_{nr} + k_p)$
Stern–Volmer	$I_f^0 / I_f = 1 + K_{SV}[Q]; \quad K_{SV} = k_q \tau_0$
HBr rate law	$d[\text{HBr}] / dt = k[\text{H}_2][\text{Br}_2]^{1/2} / (1 + k'[\text{HBr}] / [\text{Br}_2])$
HI quantum yield	$\Phi(\text{HI}) = 2$
Anthracene dimer Φ	$\Phi(A_2) = k_2[A] / (k_f + k_2[A])$

Most Frequently Asked PYQ Topics (Unit 4):

1. State the law of photochemical equivalence; give reasons why reactions deviate.
2. Define quantum yield; distinguish primary vs. overall quantum yield.
3. Derive the Stern–Volmer equation; draw and label the plot; explain how k_f , k_q , τ_0 are determined.
4. Derive the rate of HBr formation in $\text{H}_2 + \text{Br}_2$ photochemical reaction; find its quantum yield.
5. Derive the quantum yield of HI photodecomposition ($\Phi = 2$).
6. Derive the quantum yield of anthracene dimerisation; explain why it is small and increases with $[A]$.
7. Differentiate thermal equilibrium from photostationary state (table + explanation).
8. Define resonance emission; contrast with fluorescence.

9. Three deactivation processes of a photoexcited molecule with the Jablonski diagram.
10. Reasons for low and high quantum yield (at least 3 each).
11. What is chemical actinometry? Describe any two actinometers.

Central Strategy for Unit 4 Problems:

1. Write the full mechanism (initiation, propagation, inhibition, termination).
2. Apply the **photostationary-state approximation** ($d[A^*]/dt = 0$) to eliminate $[A^*]$.
3. Apply steady-state to all radical/intermediate species as needed.
4. Divide the rate of product formation by I_a to obtain the quantum yield Φ .